MTH1004

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

May 2023

Probability, Statistics, and Data

Module Leader: Dr Stefan Siegert

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Statistical tables (Murdoch & Barnes) will be provided. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No Word Count specified

SECTION A

1. (a) Among a group of fifteen injured rugby players, all have either an ankle sprain, a bruised shoulder, a concussion, or a combination of these injuries. Eight have an ankle sprain, ten have a bruised shoulder, and nine have a concussion. Further, six have both an ankle sprain and a bruised shoulder, four have both an ankle sprain and a concussion, and five have both a bruised shoulder and a concussion. How many have all three injuries? (9)

   (b) A chemical test is used to detect the presence or absence of an environmental pollutant in the soil. Before any testing is done, the pollutant has a 20% probability of being present in the soil. The chemical test is known to be 85% accurate, that is, it detects presence and absence correctly with 85% probability. What is the conditional probability that the pollutant is present in the soil, given a positive test result? (14)

   (c) The target for the mean response time of an emergency service is 10 minutes. A simple random sample of $n = 100$ response times were measured. The mean of the 100 times was $\bar{y} = 10.6$ minutes and their standard deviation was $s = 2.0$ minutes.

      (i) Conduct a test of size 5% of the null hypothesis that the population mean response time equals 10 minutes against the alternative hypothesis that the population mean response time is greater than 10 minutes. State clearly your test statistic, its null distribution, the critical region, and your conclusion.

      (ii) Compute an equal-tailed 95% confidence interval for the population mean response time. (15)

   (d) Figure 1 (on page 3) plots quantiles of a sample of monthly rainfall totals (measured in millimetres, mm) against quantiles of a normal distribution. Discuss briefly whether the location, scale and shape of the normal distribution fit the data and propose another model that might fit these data well. (6)

   (e) Some researchers seek to learn about the extent to which students at UK universities cheat in assessments. To collect their data, the researchers email all students at Exeter and ask them to complete an on-line survey containing questions such as 'Have you ever submitted for assessment work that is not your own?'. Briefly describe three sources of bias which could affect the validity of the researchers' inferences. (6)

[**50**]

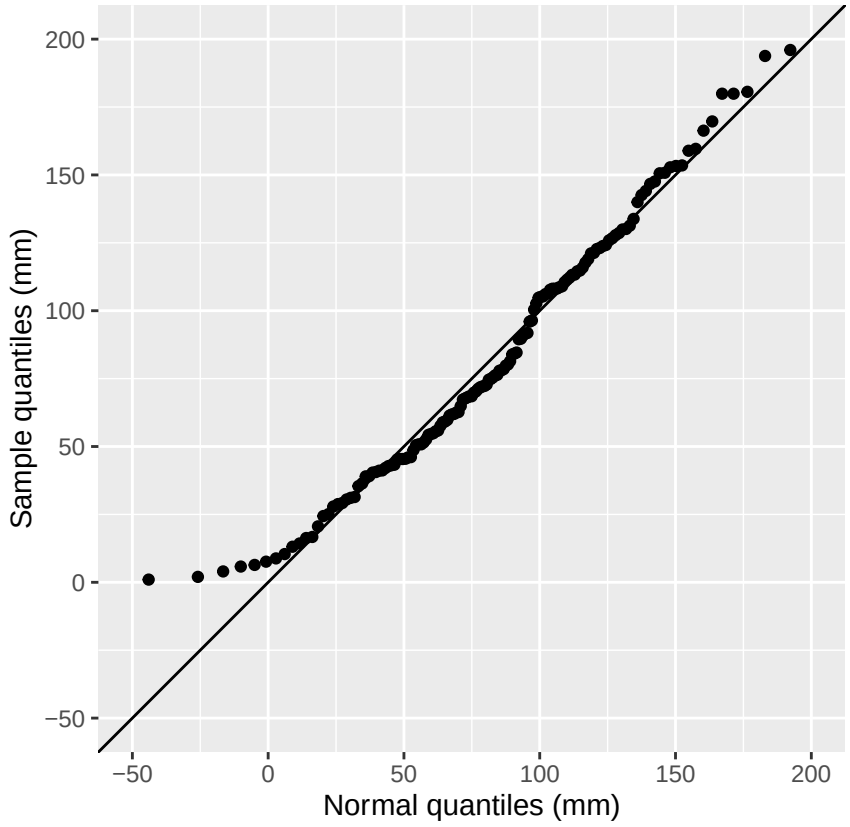


Figure 1: Used in Question 1d.

SECTION B

2. Let X_1 and X_2 be independent $\text{Exp}(\theta)$ random variables with probability density functions $f_{X_1}(x) = f_{X_2}(x) = \theta \exp(-\theta x)$ for $x > 0$. Calculate the probability density function of $Y_1 = X_1/(X_1 + X_2)$ and hence name the distribution of Y_1. **[25]**

3. (a) Let $X_1, \ldots, X_n$ be independent $\text{Unif}(0, \beta)$ random variables with probability density function $f(x) = 1/\beta$ for $0 < x < \beta$.

   (i) Calculate the method of moments estimator for β.

   (ii) Determine whether this estimator is unbiased. (10)

   (b) Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be two independent and unbiased estimators for a parameter, θ, with $\text{Var}(\hat{\theta}_1) = v_1$ and $\text{Var}(\hat{\theta}_2) = v_2$. Define

   $$\hat{\theta} = a\hat{\theta}_1 + (1 - a)\hat{\theta}_2$$

   for a real constant a.

   (i) Show that $\hat{\theta}$ is unbiased.

   (ii) Find the value of a that minimises the variance of $\hat{\theta}$. (15)

   [25]

4. Let X be a continuous random variable with range space $(-\infty, \infty)$, finite expectation, and probability density function $f(x)$. The pdf satisfies the relation $f(c + x) = f(c - x)$ for some finite constant $c \in \mathbb{R}$ and for all values of $x \in \mathbb{R}$.

   (a) Using the substitutions $y = x - c$ and $y = c - x$, or otherwise, show that

   $$\int_{-\infty}^{c} f(x)dx = \int_{c}^{\infty} f(x)dx.$$

   (5)

   (b) Hence, or otherwise, show that the median of X is equal to c. (6)

   (c) Using suitable substitutions, or otherwise, show that

   $$\int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{\infty} xf(c + x)dx + c.$$

   (6)

   (d) Hence, or otherwise, show that the expectation of X is equal to c. (8)

   [25]